Azure Lighthouse Delegated Administration Design Study

Project Tools

Course 24CC3046, Project Number P079 | My Team: T079 | 4 Students | Track: Microsoft Certified: Azure Administrator Associate

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The tools below cover identity and delegation, infrastructure-as-code, governance and monitoring, and version control needed to design, deploy, and audit the Azure Lighthouse cross-tenant model.

Identity & delegation

Tool	Purpose in this project	Cost
Azure Active Directory	Hosts provider and customer tenants; security groups receive delegated roles.	Unpaid tier used
Azure Lighthouse	Core delegation mechanism; registration definitions/assignments project scoped access into customer tenants.	Unpaid
Azure RBAC	Assigns least-privilege roles (Reader, Contributor, custom roles) at the delegated scope.	Unpaid

Infrastructure as code

Tool	Purpose in this project	Cost
ARM templates / Bicep	Defines the Lighthouse onboarding package (registration Definition + registration Assignment) each customer deploys.	Unpaid
Azure CLI / PowerShell (Az module)	Deploys templates, validates delegations, and scripts role assignment checks.	Unpaid

Governance & monitoring

Tool	Purpose in this project	Cost
Azure Policy	Enforces compliance guardrails across delegated customer subscriptions.	Unpaid (policy engine)
Azure Resource Graph	Runs cross-tenant queries for consolidated visibility into delegated resources.	Unpaid

Log Analytics workspace	Collects activity and diagnostic logs; ingestion volume is billed.	Paid (usage-based)
Microsoft Sentinel	Aggregates and correlates security signals across tenants for unified auditing.	Paid (usage-based)

Documentation & collaboration

Tool	Purpose in this project	Cost
Git / GitHub	Version control for templates, diagrams, and design documents.	Unpaid (public/free tier)
Draw.io / Visio or diagram-as-code	Produces the architecture and delegation diagrams.	Unpaid (free tiers available)

Note on cost: Azure Lighthouse itself carries no additional charge — it is a free control-plane capability. Cost only arises from the underlying Azure services it is used to manage or monitor, such as Log Analytics ingestion and Microsoft Sentinel, which are billed on usage.